

PNEUMOAI

AI-ASSISTED CHEST X-RAY ANALYSIS REPORT

DenseNet121 Deep Learning Architecture • Chest Radiograph Screening

Patient Information

Full Name: John Doe	Age: 45
Sex: Male	Contact: 9876543210
Examination: Chest X-Ray (CXR)	Date: 17 September 2026, 19:48

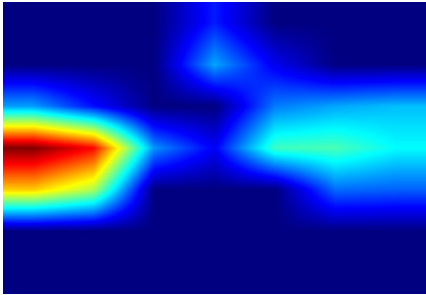
AI Analysis Results

AI Prediction: PNEUMONIA	Individual Confidence: 78.3%
Model: DenseNet121 (Preprocessed 224×224)	Classification Threshold: 0.65

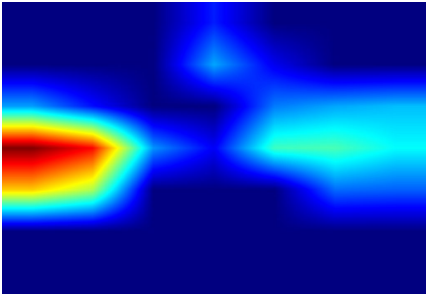
Radiographic Image Analysis & AI Attention

Grad-CAM neural activation maps highlight anatomical regions influencing the deep learning prediction.

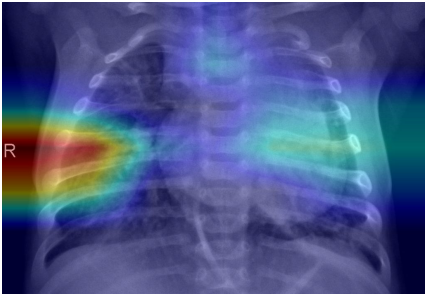
Original Chest X-Ray



AI Attention (Grad-CAM)



X-Ray + Grad-CAM Overlay



Model Test Performance Reference

Model Architecture: DenseNet121	Test Set Images: 624 chest radiographs
Test Accuracy: 91.51%	Sensitivity (Recall): 93.33%
Specificity: 88.46%	Pneumonia Precision: 93.09%
Pneumonia F1-Score: 93.21%	Confusion Matrix: [[207, 27], [26, 364]]

Note: The above metrics represent independent model evaluation performance on the test benchmark, not individual patient confidence.